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Tracking Device Operation in Safety-Classified Zone

Abstract

Tracking devices can be associated with safe zones, smart zones, and high risk zones. Safe zones correspond to regions where a likelihood that a tracking device is lost within the safe zone is lower than outside the safe zone. High risk zones correspond to regions where a likelihood that a tracking device is lost within the high risk zone is higher than outside the high risk zone. Smart zones correspond to an expected tracking device, mobile device, or user behavior. Home areas are geographic regions in which a user resides, and travel areas are geographic regions in which a user does not reside. A tracking device can be configured to operate in a mode selected based on a presence of the tracking device within a safe zone, a smart zone, a high risk zone, a home area, or a travel area.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/645,242, filed Apr. 24, 2024, which is a continuation of U.S. application Ser. No. 17/984,940, filed Nov. 10, 2022, now U.S. Pat. No. 12,004,040, which is a continuation of U.S. application Ser. No. 17/137,174, filed Dec. 29, 2020, now U.S. Pat. No. 11,533,582, which is a continuation of U.S. application Ser. No. 16/927,993, filed Jul. 14, 2020, now U.S. Pat. No. 10,911,889, which is a continuation of U.S. application Ser. No. 16/818,638, filed Mar. 13, 2020, now U.S. Pat. No. 10,764,712, which is a continuation of U.S. application Ser. No. 16/703,548, filed Dec. 4, 2019, now U.S. Pat. No. 10,631,121, which is a continuation of U.S. application Ser. No. 16/401,593, filed May 2, 2019, now U.S. Pat. No. 10,536,801, which is a continuation of U.S. application Ser. No. 16/184,252, filed Nov. 8, 2018, now U.S. Pat. No. 10,327,101, which is a continuation of U.S. application Ser. No. 16/055,811, filed Aug. 6, 2018, now U.S. Pat. No. 10,187,746, which application is a continuation of U.S. application Ser. No. 15/841,841, filed Dec. 14, 2017, now U.S. Pat. No. 10,070,255, which application claims the

benefit of U.S. Provisional Application No. 62/546,454, filed Aug. 16, 2017, all of which are incorporated by reference in their entirety.

BACKGROUND

[0002] This disclosure relates generally to tracking devices, and more specifically, to modifying the behavior of a tracking device or an associated mobile device based on its presence in a zone or an area.

[0003] Electronic tracking devices have created numerous ways for people to track the locations of people and/or objects. For example, a user can use GPS technology to track a device remotely or determine a location of the user. In another example, a user can attach a tracking device to an important object, such as keys or a wallet, and use the features of the tracking device to more quickly locate the object, (e.g., if it becomes lost).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 illustrates an example tracking system environment in which a tracking device can operate, according to one embodiment.

[0005] FIG. 2 illustrates an example tracking system for use in a tracking system environment, according to one embodiment.

[0006] FIG. 3 illustrates an example user mobile device for use in a tracking system environment, according to one embodiment.

[0007] FIG. 4 illustrates an example community mobile device for use in a tracking system environment, according to one embodiment.

[0008] FIG. 5 illustrates an example tracking device for use in a tracking system environment, according to one embodiment.

[0009] FIG. 6 illustrates a plurality of zones with associated behavior profiles that can affect the behavior of a tracking device or an associated mobile device, according to one embodiment.

[0010] FIG. 7 illustrates a plurality of areas that each contain a plurality of zones, according to one embodiment.

[0011] FIG. 8 is a flow chart illustrating a process for generating zones, according to one embodiment,

[0012] FIG. 9 is a flow chart illustrating a process for modifying tracking device behavior based on presence in a smart zone, according to one embodiment.

[0013] FIG. 10 is a flow chart illustrating a process for modifying tracking device behavior based on presence in an area, according to one embodiment.

[0014] The figures depict various embodiments of the present invention for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION

Environment Overview

[0015] Embodiments described herein detail functionality associated with a tracking device. A user can attach a tracking device to or enclose the tracking device within an object, such as a wallet, keys, a car, a bike, a pet, or any other object that the user wants to track. The user can then use a mobile device (e.g., by way of a software application installed on the mobile device) or other device or service to track the tracking device and corresponding object. For example, the mobile device can perform a local search for a tracking device attached to a near-by object. However, in situations where the user is unable to locate the tracking device using their own mobile device (e.g., if the tracking device is beyond a distance within which the mobile device and the tracking device

can communicate), the user can leverage the capabilities of a community of users of a tracking device system.

[0016] In particular, a tracking system (also referred to herein as a “cloud server” or simply “server”) can maintain user profiles associated with a plurality of users of the tracking device system. The tracking system can associate each user within the system with one or more tracking devices associated the user (e.g., tracking devices that the user has purchased and is using to track objects owned by the user). If the user's object becomes lost or stolen, the user can send an indication that the tracking device is lost to the tracking system, which is in communication with one or more mobile devices associated with the community of users in communication with the system. The tracking system can set a flag indicating the tracking device is lost. When one of a community of mobile devices that are scanning for nearby tracking devices and providing updated locations to the tracking system identifies a flagged tracking device, the tracking system can associate the received location with the flagged tracking device, and relay the location to a user of the tracking device, thereby enabling the user to locate the lost tracking device. As used herein, “mobile device” can refer to a phone, tablet computer, or other connected device, and can also refer to systems typically not consider mobile, such as servers, routers, gateways, access points, and specialized systems configured to couple to tracking devices and report a location of the tracking devices.

[0017] As used herein, “tracking device” can refer to any device configured to communicate with another device for the purpose of locating the tracking device. Tracking devices can be specialized or single-purpose devices (e.g., self-contained devices that include circuitry or components to communicate with another device). However, “tracking device” as used herein can also refer to device or object with a different primary function but with secondary tracking device functionality. For example, a suitcase can include tracking device components that allow a user to track and/or locate the suitcase. In some embodiments, a tracking device platform can be established such that devices and objects that satisfy one or more criteria can act as tracking devices within a tracking device ecosystem. For instance, a tracking device provider can provide an SDK or custom chipset that, when incorporated into an object or device, enable the object or device to function as tracking devices, to communicate with other devices within the tracking device ecosystem, and to implement the functionalities described herein.

[0018] FIG. 1 illustrates an example tracking system environment in which a tracking device can operate, according to one embodiment. The environment of FIG. 1 includes a tracking system **100** communicatively coupled to a mobile device **102** associated with the user **103** via a first network **108**. The tracking system **100** is also communicatively coupled to a plurality of community mobile devices **104a** through **104n** (collectively referred to herein as “community mobile devices **104**”) associated with a plurality of users **105a** through **105n** of the tracking system **100** (collectively referred to herein as “community users **105**”) via the first network **108**. As will be explained in more detail below, the tracking system **100** can allow the user **103** to manage and/or locate a tracking device **106** associated with the user **103**. In some embodiments, the tracking system **100** leverages the capabilities of community mobile devices **104** to locate the tracking device **106** if the location of the tracking device is unknown to the user **103** and beyond the capabilities of mobile device **102** to track. In some configurations, the user **103** may own and register multiple tracking devices **106**. Although FIG. 1 illustrates a particular arrangement of the tracking system **100**, mobile device **102**, community mobile devices **104**, and tracking device **106**, various additional arrangements are possible.

[0019] In some configurations, the user **103** may be part of the community of users **105**. Further, one or more users **105** may own and register one or more tracking devices **106**. Thus, any one of the users within the community of users **105** can communicate with tracking system **100** and leverage the capabilities of the community of users **105** in addition to the user **103** to locate a tracking device **106** that has been lost.

[0020] The tracking system **100**, mobile device **102**, and plurality of community mobile devices **104** may communicate using any communication platforms and technologies suitable for transporting data and/or communication signals, including known communication technologies, devices, media, and protocols supportive of remote data communications.

[0021] In certain embodiments, the tracking system **100**, mobile device **102**, and community mobile devices **104** may communicate via a network **108**, which may include one or more networks, including, but not limited to, wireless networks (e.g., wireless communication networks), mobile telephone networks (e.g., cellular telephone networks), closed communication networks, open communication networks, satellite networks, navigation networks, broadband networks, narrowband networks, the Internet, local area networks, and any other networks capable of carrying data and/or communications signals between the tracking system **100**, mobile device **102**, and community mobile devices **104**. The mobile device **102** and community of mobile devices **104** may also be in communication with a tracking device **106** via a second network **110**. The second network **110** may be a similar or different type of network as the first network **108**. In some embodiments, the second network **110** comprises a wireless network with a limited communication range, such as a Bluetooth or Bluetooth Low Energy (BLE) wireless network. In some configurations, the second network **110** is a point-to-point network including the tracking device **106** and one or more mobile devices that fall within a proximity of the tracking device **106**. In such embodiments, the mobile device **102** and community mobile devices **104** may only be able to communicate with the tracking device **106** if they are within a close proximity to the tracking device, though in other embodiments, the tracking device can use long-distance communication functionality (for instance, a GSM transceiver) to communicate with either a mobile device **102/104** or the tracking system **100** at any distance. In some configurations, the mobile device **102** and one or more community mobile devices **104** may each be associated with multiple tracking devices associated with various users.

[0022] As mentioned above, FIG. **1** illustrates the mobile device **102** associated with the user **103**. The mobile device **102** can be configured to perform one or more functions described herein with respect to locating tracking devices (e.g., tracking device **106**). For example, the mobile device **102** can receive input from the user **103** representative of information about the user **103** and information about a tracking device **106**. The mobile device **102** may then provide the received user information, tracking device information, and/or information about the mobile device **102** to the tracking system **100**. Accordingly, the tracking system **100** is able to associate the mobile device **102**, the user **103**, and/or the tracking device **106** with one another. In some embodiments, the mobile device **102** can communicate with the tracking device **106** and provide information regarding the location of the tracking device to the user **103**. For example, the mobile device **102** can detect a communication signal from the tracking device **106** (e.g., by way of second network **110**) as well as a strength of the communication signal or other measure of proximity to determine an approximate distance between the mobile device **102** and the tracking device **106**. The mobile device **102** can then provide this information to the user **103** (e.g., by way of one or more graphical user interfaces) to assist the user **103** to locate the tracking device **106**. Accordingly, the user **103** can use the mobile device **102** to track and locate the tracking device **106** and a corresponding object associated with the tracking device **106**. If the mobile device **102** is located beyond the immediate range of communication with the tracking device **106** (e.g., beyond the second network **110**), the mobile device **102** can be configured to send an indication that a tracking device **106** is lost to the tracking system **100**, requesting assistance in finding the tracking device. The mobile device **102** can send an indication of a lost device in response to a command from the user **103**. For example, once the user **103** has determined that the tracking device **106** is lost, the user can provide user input to the mobile device **102** (e.g., by way of a graphical user interface), requesting that the mobile device **102** send an indication that the tracking device **106** is lost to the tracking system **100**. In some examples, the lost indication can include information identifying the user **103** (e.g.,

name, username, authentication information), information associated with the mobile device **102** (e.g., a mobile phone number), information associated with the tracking device (e.g., a unique tracking device identifier), or a location of the user (e.g., a GPS location of the mobile device **102** at the time the request is sent).

[0023] The tracking system **100** can be configured to provide a number of features and services associated with the tracking and management of a plurality of tracking devices and/or users associated with the tracking devices. For example, the tracking system **100** can manage information and/or user profiles associated with user **103** and community users **105**. In particular, the tracking system **100** can manage information associated with the tracking device **106** and/or other tracking devices associated with the user **103** and/or the community users **105**.

[0024] As mentioned above, the tracking system **100** can receive an indication that the tracking device **106** is lost from the mobile device **102**. The tracking system **100** can then process the indication in order to help the user **103** find the tracking device **106**. For example, the tracking system **100** can leverage the capabilities of the community mobile devices **104** to help find the tracking device **106**. In particular, the tracking system **100** may set a flag for a tracking device **106** to indicate that the tracking device **106** is lost and monitor communications received from the community mobile devices **104** indicating the location of one or more tracking devices **106** within proximity of the community mobile devices **104**. The tracking system **100** can determine whether a specific location is associated with the lost tracking device **106** and provide any location updates associated with the tracking device **106** to the mobile device **102**. In one example, the tracking system may receive constant updates of tracking device **106** locations regardless of whether a tracking device **106** is lost and provide a most recent updated location of the tracking device **106** in response to receiving an indication that the tracking device **106** is lost.

[0025] In some configurations, the tracking system **100** can send a location request associated with the tracking device **106** to each of the community mobile devices **104**. The location request can include any instructions and/or information necessary for the community mobile devices **106** to find the tracking device **102**. For example, the location request can include a unique identifier associated with the tracking device **106** that can be used by the community mobile devices **104** to identify the tracking device **106**. Accordingly, if one of the community mobile devices **104** detects a communication from the tracking device **106** (e.g., if the community mobile device **104** is within range or moves within range of the communication capabilities of the tracking device **106** and receives a signal from the tracking device **106** including or associated with the unique identifier associated with the tracking device **106**), the community mobile device **104** can inform the tracking system **100**. Using the information received from the community mobile devices **104**, the tracking system **100** can inform the user (e.g., by way of the mobile device **102**) of a potential location of the tracking device **106**.

[0026] As shown in FIG. **1** and as mentioned above, the tracking system **100** can communicate with a plurality of community mobile devices **104** associated with corresponding community users **105**. For example, an implementation may include a first community mobile device **104a** associated with a first community user **105a**, a second community mobile device **104b** associated with a second community user **105b**, and additional communication mobile devices associated with additional community users up to an nth community mobile device **104n** associated with an nth community user **105n**. The community mobile devices **104** may also include functionality that enables each community mobile device **104** to identify a tracking device **106** within a proximity of the community mobile device **104**. In one example, a first community mobile device **104a** within proximity of a tracking device **106** can communicate with the tracking device **106**, identify the tracking device **106** (e.g., using a unique identifier associated with the tracking device **106**), and/or detect a location associated with the tracking device **106** (e.g., a location of the first mobile community device **104a** at the time of the communication with the tracking device **106**). This information can be used to provide updated locations and/or respond to a location request from the

tracking system **100** regarding the tracking device **106**. In some embodiments, the steps performed by the first community mobile device **104a** can be hidden from the first community user **105a**. Accordingly, the first community mobile device **104a** can assist in locating the tracking device **106** without bother and without the knowledge of the first community user **105a**.

[0027] As mentioned above, the tracking system **100** can assist a user **103** in locating a tracking device **106**. The tracking device may be a chip, tile, tag, or other device for housing circuitry and that may be attached to or enclosed within an object such as a wallet, keys, purse, car, or other object that the user **103** may track. Additionally, the tracking device **106** may include a speaker for emitting a sound and/or a transmitter for broadcasting a beacon. In one configuration, the tracking device **106** may periodically transmit a beacon signal that may be detected using a nearby mobile device **102** and/or community mobile device **104**. In some configurations, the tracking device **106** broadcasts a beacon at regular intervals (e.g., one second intervals) that may be detected from a nearby mobile device (e.g., community mobile device **104**). The strength of the signal emitted from the tracking device **106** may be used to determine a degree of proximity to the mobile device **102** or community mobile device **104** that detects the signal. For example, a higher strength signal would indicate a close proximity between the tracking device **106** and the mobile device **102** and a lower strength signal would indicate a more remote proximity between the tracking device **106** and the mobile device **102**, though in some embodiments, the tracking device **106** can intentionally vary the transmission strength of the beacon signal. In some cases, the strength of signal or absence of a signal may be used to indicate that a tracking device **106** is lost.

System Overview

[0028] FIG. 2 illustrates an example tracking system for use in a tracking system environment, according to one embodiment. As shown, the tracking system **100** may include, but is not limited to, an association manager **204**, a tracking device location manager **206**, a zone manager **207A**, an area manager **207B**, and a data manager **208**, each of which may be in communication with one another using any suitable communication technologies. It will be recognized that although managers **204-208** are shown to be separate in FIG. 2, any of the managers **204-208** may be combined into fewer managers, such as into a single manager, or divided into more managers as may serve a particular embodiment.

[0029] The association manager **204** may be configured to receive, transmit, obtain, and/or update information about a user **103** and/or information about one or more specific tracking devices (e.g., tracking device **106**). In some configurations, the association manager **204** may associate information associated with a user **103** with information associated with a tracking device **106**. For example, user information and tracking information may be obtained by way of a mobile device **102**, and the association manager **204** may be used to link the user information and tracking information. The association between user **103** and tracking device **106** may be used for authentication purposes, or for storing user information, tracking device information, permissions, or other information about a user **103** and/or tracking device **106** in a database.

[0030] The tracking system **100** also includes a tracking device location manager **206**. The tracking device location manager **206** may receive and process an indication that the tracking device **106** is lost from a mobile device (e.g., mobile device **102** or community mobile devices **104**). For example, the tracking system **100** may receive a lost indication from a mobile device **102** indicating that the tracking device **106** is lost. The tracking device location manager **206** may set a flag on a database (e.g., tracker database **212**) indicating that the tracking device **106** is lost. The tracking device location manager **206** may also query a database to determine tracking information corresponding to the associated user **103** and/or tracking device **106**. The tracking system **100** may obtain tracking device information and provide the tracking device information or other information associated with the tracking device **106** to a plurality of community mobile devices **104** to be on alert for the lost or unavailable tracking device **106**.

[0031] The tracking device location manager **206** may also receive a location from one or more

community mobile devices **104** that detect the tracking device **106**, for instance in response to the community mobile device receiving a beacon signal transmitted by the tracking device **106**, without the tracking device **106** having been previously marked as lost. In such embodiments, a user corresponding to the mobile device **102** can request a most recent location associated with the tracking device from the tracking system **100**, and the location manager **206** can provide the location received from the community mobile device for display by the mobile device **102**. In some embodiments, the location manager **206** provides the location of the tracking device **106** received from a community mobile device either automatically (for instance if the tracking device **106** is marked as lost) or at the request of a user of the mobile device **102** (for instance, via an application on the mobile device **102**). The location manager **206** can provide a location of a tracking device **106** to a mobile device **102** via a text message, push notification, application notification, automated voice message, or any other suitable form of communication.

[0032] The tracking device location manager **206** may further manage providing indications about whether a tracking device **106** is lost or not lost. For example, as discussed above, the tracking device location manager **206** may provide a location request to the community of mobile devices **104** indicating that a tracking device **106** is lost. Additionally, upon location of the tracking device **106** by the user **103** or by one of the community of users **105**, the tracking device location manager **206** may provide an indication to the user **103**, community user **105**, or tracking system **100** that the tracking device **106** has been found, thus removing any flags associated with a tracking device and/or canceling any location request previously provided to the community of users **105**. For example, where a user **103** sends an indication that the tracking device **106** is lost to the tracking system **100** and later finds the tracking device **106**, the mobile device **102** may provide an indication to the tracking system **100** that the tracking device **106** has been found. In response, the tracking device location manager **206** may remove a flag indicating that the tracking device **106** is lost and/or provide an updated indication to the community of users **105** that the tracking device **106** has been found, thus canceling any instructions associated with the previously provided location request. In some configurations, the notification that the tracking device **106** has been found may be provided automatically upon the mobile device **102** detecting the tracking device **106** within a proximity of the mobile device **102**. Alternatively, the notification that the tracking device **106** has been found may be provided by the user **103** via user input on the mobile device **102**. In another example, a known user (e.g., a friend or family member) with whom the tracking device **106** has been shared may provide an indication that the tracking device **106** has been found.

[0033] The zone manager **207A** generates different types of zones based on a location dataset for one or more users. After generating a zone, the zone manager **207A** stores the zone in the zone database. The generation and functionality of zones is described in further detail below with reference to FIG. 6. As described below, some types of zones (e.g., smart zones and safe zones) are specific to a user, whereas other types of zones (e.g., high-risk zones) are applicable to multiple users. In one embodiment, the zone manager **207A** stores user-specific zones in the user database **210** in association with other data about the user, and the zone manager **207A** stores zones that are applicable to multiple users in the zone database **218**. In another embodiment, the zone manager **207A** stores every generated zone in the zone database **218**, but user-specific zones are stored with an identifier for the associated user.

[0034] The area manager **207B** generates different types of areas and stores generated areas in the area database **220**. The generation and functionality of areas is described in further detail below with reference to FIG. 7. As described below, areas may be user specific. Thus, similar to user-specific zones, user-specific areas may be stored in the user database **210** in association with other data about the user, or they may be stored in the area database **220** with an identifier for the associated user.

[0035] The tracking system **100** additionally includes a data manager **208**. The data manager **208** may store and manage information associated with users, mobile devices, tracking devices,

permissions, location requests, and other data that may be stored and/or maintained in a database related to performing location services of tracking devices. As shown, the data manager **208** may include, but is not limited to, a user database **210**, a tracker database **212**, permissions data **214**, location request data **216**, a zone database, and an area database **220**. It will be recognized that although databases and data within the data manager **208** are shown to be separate in FIG. 2, any of the user database **210**, tracker database **212**, permissions data **214**, and location request data **216** may be combined in a single database or manager, or divided into more databases or managers as may serve a particular embodiment.

[0036] The data manager **208** may include the user database **210**. The user database **210** may be used to store data related to various users. For example, the user database **210** may include data about the user **103** as well as data about each user **105** in a community of users **105**. The community of users **105** may include any user that has provided user information to the tracking system **100** via a mobile device **102**, **104** or other electronic device. The user information may be associated with one or more respective tracking devices **106**, or may be stored without an association to a particular tracking device. For example, a community user **105** may provide user information and permit performance of tracking functions on the community mobile device **104** without owning or being associated with a tracking device **106**. The user database **210** may also include information about one or more mobile devices or other electronic devices associated with a particular user.

[0037] The data manager **208** may also include a tracker database **212**. The tracker database **212** may be used to store data related to tracking devices. For example, the tracker database **212** may include tracking data for any tracking device **106** that has been registered with the tracking system **100**. Tracking data may include unique tracker identifications (IDs) associated with individual tracking devices **106**. Tracker IDs may be associated with a respective user **103**. Tracker IDs may also be associated with multiple users. Additionally, the tracker database **212** may include any flags or other indications associated with whether a specific tracking device **106** has been indicated as lost and whether any incoming communications with regard to that tracking device **106** should be processed based on the presence of a flag associated with the tracking device **106**.

[0038] The data manager **208** may further include permissions data **214** and location request data **216**. Permissions data **214** may include levels of permissions associated with a particular user **103** and/or tracking device **106**. For example, permissions data **214** may include additional users that have been indicated as sharing a tracking device **106**, or who have been given permission to locate or receive a location of a tracking device **106**. Location request data **216** may include information related to a location request or a lost indication received from the user **103** via a mobile device **102**.

[0039] The data manager **208** may additionally include a zone database **218** and an area database **220**. As described above, the zone database **218** may store zones generated by the zone manager **207A**, and the area database **220** may store areas generated by the area manager **207B**.

[0040] FIG. 3 illustrates an example user mobile device for use in a tracking system environment, according to one embodiment. As shown, the mobile device **102** may include, but is not limited to, a user interface manager **302**, a location request manager **304**, a database manager **306**, and a tracking manager **308**, each of which may be in communication with one another using any suitable communication technologies. It will be recognized that although managers **302-308** are shown to be separate in FIG. 3, any of the managers **302-308** may be combined into fewer managers, such as into a single manager, or divided into more managers as may serve a particular embodiment.

[0041] As will be explained in more detail below, the mobile device **102** includes the user interface manager **302**. The user interface manager **302** may facilitate providing the user **103** access to data on a tracking system **100** and/or providing data to the tracking system **100**. Further, the user interface manager **302** provides a user interface by which the user **103** may communicate with tracking system **100** and/or tracking device **106** via mobile device **102**.

[0042] The mobile device **102** may also include a location request manager **304**. The location

request manager **304** may receive and process a request input to the mobile device **102** to send an indication that a tracking device **106** is lost to a tracking system **100**. For example, the user **103** may provide an indication that a tracking device **106** is lost, unreachable, or otherwise unavailable from the mobile device **102** via the user interface manager **302**, and the location request manager **304** may process the lost indication and provide any necessary data to the tracking system **100** for processing and relaying a location request to other users **105** over a network **108**. In some configurations, an indication that a tracking device **106** is lost is provided via user input. Alternatively, the indication may be transmitted automatically in response to the mobile device **102** determining that a tracking device **106** is lost.

[0043] In addition, the location request manager **304** can request a location of the tracking device **106** without the tracking device **106** being identified as lost. For instance, a user can access a tracking device location feature of an application running on the mobile device **102** (for example, via the user interface manager **302**), and the location request manager **304** can request a most recent location of the tracking device **106** from the tracking system **100**. The location request manager **304** can receive the most recent location from the tracking system **100**, and can display the most recent location via the user interface manager **302**.

[0044] The mobile device **102** may also include a database manager **306**. The database manager **306** may maintain data related to the user **103**, tracking device **106**, permissions, or other data that may be used for locating a tracking device **106** and/or providing a request to a tracking system **100** for locating one or more tracking devices **106** associated with the user **103**. Further, the database manager **306** may maintain any information that may be accessed using any other manager on the mobile device **102**.

[0045] The mobile device **102** may further include a tracking manager **308**. The tracking manager **308** may include a tracking application (e.g., a software application) for communicating with and locating a tracking device **106** associated with the user **103**. For example, the tracking manager **308** may be one configuration of a tracking application installed on the mobile device **102** that provides the functionality for locating a tracking device **106** and/or requesting location of a tracking device **106** using a tracking system **100** and/or a plurality of community mobile devices **104**. As shown, the tracking manager **308** may include, but is not limited to, a Bluetooth Low Energy (BLE) manager **310**, a persistence manager **312**, a local files manager **314**, a motion manager **316**, a secure storage manager **318**, a settings manager **320**, a location manager **322**, a network manager **324**, a notification manager **326**, a sound manager **328**, a friends manager **330**, a photo manager **332**, an authentication manager **334**, and a device manager **336**. Thus, the tracking manager **308** may perform any of the functions associated with managers **310-338**, described in additional detail below.

[0046] The BLE manager **310** may be used to manage communication with one or more tracking devices **106**. The persistence manager **312** may be used to store logical schema information that is relevant to the tracking manager **308**. The local files manager **314** may be responsible for managing all files that are input or output from the mobile device **102**. The motion manager **316** may be responsible for all motion management required by the tracking manager **308**. The secure storage manager may be responsible for storage of secure data, including information such as passwords and private data that would be accessed through this sub-system. The settings manager **320** may be responsible for managing settings used by the tracking manager **308**. Such settings may be user controlled (e.g., user settings) or defined by the tracking manager **308** for internal use (e.g., application settings) by a mobile device **102** and/or the tracking system **100**. The location manager **322** may be responsible for all location tracking done by the tracking manager **308**. For example, the location manager **322** may manage access to the location services of the mobile device **102** and works in conjunction with other managers to persist data. The network manager **324** may be responsible for all Internet communications from the tracking manager **308**. For example, the network manager **324** may mediate all Internet API calls for the tracking manager **308**. The

notification manager **326** may be responsible for managing local and push notifications required by the tracking manager **308**. The sound manager **328** may be responsible for playback of audio cues by the tracking manager **308**. The friends manager **330** may be responsible for managing access to contacts and the user's social graph. The photo manager **332** may be responsible for capturing and managing photos used by the tracking manager **308**. The authentication manager **334** may be responsible for handling the authentication (e.g., sign in or login) of users. The authentication manager **334** may also include registration (e.g., sign up) functionality. The authentication manager **334** further coordinates with other managers to achieve registration functionality. The device manager **336** may be responsible for managing the devices discovered by the tracking manager **308**. The device manager **336** may further store and/or maintain the logic for algorithms related to device discovery and update.

[0047] FIG. 4 illustrates an example community mobile device for use in a tracking system environment, according to one embodiment. As shown, the community mobile device **104** may include, but is not limited to, a user interface manager **402**, a tracking device manager **404**, a database manager **406**, and a tracking manager **408**, each of which may be in communication with one another using any suitable communication technologies. The user interface manager **402**, database manager **406**, and tracking manager **408** illustrated in FIG. 4 may include similar features and functionality as the user interface manager **302**, database manager **306**, and tracking manager **308** described above in connection with FIG. 3. It will be recognized that although managers **402-408** are shown to be separate in FIG. 4, any of the managers **402-408** may be combined into fewer managers, such as into a single manager, or divided into more managers as may serve a particular embodiment.

[0048] The community mobile device **104** may include a tracking device manager **404**. The tracking device manager **404** may facilitate scanning for nearby tracking devices **106**. In some configurations, the tracking device manager **404** can continuously or periodically scan (e.g., once per second) for nearby tracking devices **106**. The tracking device manager **404** may determine whether to provide an updated location of the nearby tracking device **106** to the tracking system **100**. In some configurations, the tracking device manager **404** provides a location of a nearby tracking device **106** automatically. Alternatively, the tracking device manager **404** may determine whether the location of the tracking device **106** has been recently updated, and may determine whether to provide an updated location based on the last time a location of the tracking device **106** has been updated (e.g., by the community mobile device **104**). For example, where the community mobile device **104** has provided a recent update of the location of a tracking device **106**, the tracking device manager **404** may decide to wait a predetermined period of time (e.g., 5 minutes) before providing an updated location of the same tracking device **106**.

[0049] In one configuration, the tracking device manager **404** may receive and process a location request or other information relayed to the community mobile device **104** by the tracking system **100**. For example, the tracking device manager **404** may receive an indication of a tracking device **106** that has been indicated as lost, and provide a location of the tracking device **106** if it comes within proximity of the community mobile device **104**. In some configurations, the community mobile device **104** is constantly scanning nearby areas to determine if there is a tracking device **106** within a proximity of the community mobile device **104**. Therefore, where a tracking device **106** that matches information provided by the tracking system **100** (e.g., from the location request) comes within proximity of the community mobile device **104**, the tracking device manager **404** may generate and transmit a response to the location request to the tracking system **100**, which may be provided to the user **103** associated with the tracking device **106**. Further, generating and transmitting the response to the tracking request may be conditioned on the status of the tracking device **106** being flagged as lost by the mobile device **102** and/or the tracking system **100**.

[0050] The tracking device manager **404** may additionally provide other information to the tracking system **100** in response to receiving the tracking request. For example, in addition to providing a

location of the community mobile device **104**, the tracking device manager may provide a signal strength associated with the location to indicate a level of proximity to the location of the community mobile device **104** provided to the user **103**. For example, if a signal strength is high, the location provided to the user **103** is likely to be more accurate than a location accompanied by a low signal strength. This may provide additional information that the user **103** may find useful in determining the precise location of tracking device **106**.

[0051] As described above, the tracking device manager **404** may determine whether to send a location within the proximity of the tracking device **106** to the tracking system **100**. The determination of whether to send a location to the tracking system **100** may be based on a variety of factors. For example, a tracking device manager **404** may determine to send a location of the tracking device **106** to a tracking system **100** based on whether the detected tracking device **106** has been indicated as lost or if a tracking request has been provided to the community mobile device **104** for the particular tracking device **106**. In some configurations, the community mobile device **104** may send an update of a location of a tracking device **106** even if the tracking device **106** is not associated with a current tracking request or if the tracking device **106** is not indicated as lost. For example, where the location of a tracking device **106** has not been updated for a predetermined period of time, the community mobile device **104** may provide an update of a tracking device location to the tracking system **100**, regardless of whether a tracking request has been received.

[0052] In some configurations, the community mobile device **104** may include additional features. For example, the community mobile device **104** may allow a tracking system **100** to snap and download a photo using photo functionality of the community mobile device **104**. In some configurations, this may be an opt-in feature by which a community user **105** permits a tracking system **100** to take a snap-shot and possibly provide a visual image of an area within a proximity of the tracking device **106**.

[0053] FIG. 5 illustrates an example tracking device for use in a tracking system environment, according to one embodiment. The tracking device **106** of FIG. 5 includes an interface **502**, a transceiver **504**, a controller **506**, one or more sensors **508**, and a GPS unit **510**. The transceiver **504** is a hardware circuit capable of both transmitting and receiving signals. It should be noted that in other embodiments, the tracking device **106** includes fewer, additional, or different components than those illustrated in FIG. 5. For instance, tracking devices might not include the GPS unit **510** and can still implement the functionalities described herein.

[0054] The interface **502** provides a communicative interface between the tracking device **106** and one or more other devices, such as a mobile device **102**. For instance, the interface **502** can instruct the transceiver **504** to output beacon signals as described above (for example, periodically or in response to a triggering event, such as a detected movement of the tracking device **106**). The interface **502** can, in response to the receiving of signals by the transceiver **504** from, for instance, the mobile device **102**, manage a pairing protocol to establish a communicative connection between the tracking device **106** and the mobile device **102**. As noted above, the pairing protocol can be a BLE connection, though in other embodiments, the interface **502** can manage other suitable wireless connection protocols (such as WiFi, Global System for Mobile Communications or GSM, and the like).

[0055] The controller **506** is a hardware chip that configures the tracking device **106** to perform one or more functions or to operate in one or operating modes or states. For instance, the controller **506** can configure the interval at which the transceiver broadcasts beacon signals, can authorize or prevent particular devices from pairing with the tracking device **106** based on information received from the devices and permissions stored at the tracking device, can increase or decrease the transmission strength of signals broadcasted by the transceiver, can configure the interface to emit a ringtone or flash an LED light, can enable or disable various tracking device sensors, can enable or disable a tracking device GPS unit, can enable or disable communicative functionality of the tracking device **106** (such as a GSM transmitter and receiving), can configure the tracking device

into a sleep mode or awake mode, can configure the tracking device into a power saving mode, and the like. The controller **506** can configure the tracking device to perform functions or to operate in a particular operating mode based on information or signals received from a device paired with or attempting to pair with the tracking device **106**, based on an operating state or connection state of the tracking device **106**, based on user-selected settings, based on information stored at the tracking device **106**, based on a detected location of the tracking device **106**, based on historical behavior of the tracking device **106** (such as a previous length of time the tracking device was configured to operate in a particular mode), based on information received from the sensors **508** or the GPS **510**, or based on any other suitable criteria.

[0056] The sensors **508** can include motion sensors (such as gyroscopes or accelerators), altimeters, orientation sensors, proximity sensors, light sensors, or any other suitable sensor configured to detect an environment of the tracking device **106**, a state of the tracking device **106**, a movement or location of the tracking device **106**, and the like. The sensors **508** are configured to provide information detected by the sensors to the controller **506**. The GPS unit **510** is configured to detect a location of the tracking device **106** based on received GPS signals, and is configured to provide detected locations to the controller **506**.

Modification of Tracking Behavior Based on Presence in Zones or Areas

[0057] FIG. **6** illustrates a plurality of zones **602**, **604**, **606** with associated behavior profiles that can affect the behavior of a tracking device or an associated mobile device, according to one embodiment. As referred to herein, a zone is a location in which an attached object has an expected behavior or a higher- or lower-than average probability of being lost or misplaced. An attached object refers to an object to which a tracking device **106** is attached. For example, an attached object may be a personal belonging that the user typically carries around with him or her, such as a set of keys, a portable electronic device (e.g., a smartphone, tablet computer, or laptop computer), a purse, or a wallet. An attached object may also be a personal belonging that is not necessarily carried around by the user on a day-to-day basis, such as a remote control or a pet. For clarity of description, the term “personal belonging” is used throughout this disclosure; when this term is used, it should be understood to refer to an object with a tracking device **106** attached to it.

[0058] Examples of three different types of zones **602**, **604**, **606** are illustrated in FIG. **6**. A first type of zone is a safe zone **602**. A safe zone is a location in which a user's probability of losing or misplacing a personal belonging is lower than average. For example, a user's home, workplace, or school may be designated as safe zones. In the example shown in FIG. **6**, the user's house has been designated as a safe zone **602**.

[0059] When one or more the user, the tracking device, or the mobile device are present in a safe zone **602** (e.g., as indicated by location tracking features of the mobile device **102**), the behavior or configuration of the user's mobile device **102** and tracking device **106** may be modified to reflect a lower probability of losing personal belongings. For instance, after detecting the user's presence in safe zone **602**, the mobile device **102** may automatically configure itself to scan for tracking device **106** less frequently. The mobile device **106** may also configure an attached tracking device **106** to enable power conservation features, such as reducing the power and frequency of the beacon signal transmitted by the tracking device **106**, reducing the frequency at which the tracking device **106** automatically transitions from sleep mode to awake mode, or reducing the volume of a speaker of the tracking device **106**. The modification of the behavior or configuration of a mobile device or tracking device when located within a safe zone is referred to herein as operation in a “safe zone mode”.

[0060] A second type of zone is a smart zone **604**. A smart zone **604** is a location in which a user and his or her personal belongings are expected to behave in a particular way. More particularly, a smart zone is a location in which the separation distance between the mobile device **102** (which is presumed to represent the location of the user) and a tracking device **106** (which is presumed to represent the location of one of the user's personal belongings) is expected to follow a particular

pattern. For example, while the user is in a restaurant, the user may leave his or her belongings at the table when the user gets up to use the restroom or speak to a waiter. Thus, in a restaurant, a temporary increase in the separation distance between the mobile device and a tracking device is likely to be the result of normal user behavior rather than an indication that the user has lost or misplaced the object attached to the tracking device.

[0061] A smart zone **604** may be associated with a behavior profile that specifies parameters governing the operation of the tracking device **106** and the associated features of the mobile device **102** when the user is present in the corresponding smart zone **604**. One parameter in the behavior profile can be, for example, the delay between detecting a disconnection of the tracking device **106** and providing a notification on the mobile device **102** that the tracking device **106** has been disconnected (hereinafter referred to as a notification delay time). Another parameter may be the threshold separation distance between the tracking device **106** and the mobile device **102** that causes the mobile device **102** to provide a notification to the user. For example, the mobile device **102** may determine a separation distance between the mobile device **102** and the tracking device **106** based on a strength of a signal received from the tracking device **106**, and if the separation distance exceeds the threshold, then the mobile device **102** provides a notification to the user in the event that the user has left the attached object behind.

[0062] Parameters specified in a behavior profile may further include parameters governing the intensity of a notification provided by the mobile device **102** or the intensity of an alert provided by the tracking device **106**. For example, the parameters may define various aspects of how a notification is provided on the mobile device **102**, such as the volume of an audio alert, the intensity or pattern of a vibration, or whether to play an audio alert, a vibration, or both. Similarly, the parameters may define aspects of how the tracking device **106** provides an alert. For instance, a parameter may define the volume of an audio alert that the tracking device **106** plays back on a built-in speaker. As another example, the tracking device **106** may provide a visual alert by causing a light on the tracking device **106** to blink, and parameters may define the frequency or brightness of the visual alert.

[0063] A behavior profile may also include parameters for power conservation features of the tracking device **106**, such as the power and frequency of the beacon signal transmitted by the tracking device **106**, the frequency at which the tracking device **106** automatically transitions from sleep mode to awake mode, or the volume of a speaker of the tracking device **106**. The behavior profile may also include parameters for features of the mobile device **102** or community mobile devices **104** in the smart zone, such as the frequency at which devices **102**, **104** in the smart zone scan for nearby tracking devices **106**, or whether to activate a search party function (e.g., configure community mobile devices **104** in the smart zone to scan for tracking devices **106** more frequently when a tracking device **106** is reported lost in the smart zone).

[0064] In one embodiment, the behavior profile for a smart zone is stored on the tracking system **100** (e.g., in association with the smart zone in the zone database **218**, or in association with the user and the smart zone in the user database **210**). Behavior profiles may additionally or alternatively be stored on the user's mobile device **102**. For example, behavior profiles are stored on the tracking system **110**, and the mobile device **102** maintains a cache containing copies of behavior profiles and associated smart zones for locations that the user frequently visits (e.g., more than a threshold number of times in a trailing time period, such as the preceding week or month).

[0065] A smart zone **604** may also be associated with a category representing the type of real-world location corresponding to the smart zone **604** (e.g., a restaurant, gym, church, etc.). For example, the smart zone **604A** is categorized as a restaurant smart zone, and the smart zone **604B** is categorized as a gym smart zone. An example process for generating smart zones and assigning categories to smart zones is described below with reference to FIG. 7A.

[0066] Each smart zone category may be associated with a behavior profile specifying parameters that reflect the expected behavior of users, mobile devices, or tracking devices in the corresponding

type of location. For example, a restaurant behavior profile (e.g., for the restaurant smart zone **604A**) specifies a notification delay time that is relatively short, but greater than 0 minutes (e.g., between 15 and 30 minutes) as long as the location of the mobile device **102** remains inside the restaurant smart zone. As noted above, when dining at a restaurant, it is common for a user to leave his or her belongings at the table for short periods of time in order to use the restroom, visit a buffet, or speak to a waiter. Thus, a short but nonzero notification delay time in a restaurant smart zone prevents the mobile device **102** from providing unnecessary notifications that a tracking device **106** may have been lost when the user has simply left the table for a short period of time. [0067] In one embodiment, the delay time for every restaurant is the same default value (e.g., between 15 and 30 minutes). In some embodiments, the delay time for a particular restaurant smart zone can be generated (or adjusted from the default value) based on tracking device data received from a plurality of users who have visited the restaurant. In one embodiment, the tracking system **100** computes an average disconnect-to-reconnect time for tracking devices **106** in the restaurant (e.g., an average length of time between a tracking device **106** disconnecting from a mobile device **102** and the tracking device **106** reconnecting to the mobile device) and generates the delay time for the restaurant smart zone based on the average disconnect-to-reconnect time (e.g., by multiplying the disconnect-to-reconnect time by a predetermined scale factor, such as 2.0). Because the average disconnect-to-reconnect time can represent the average time a user leaves the table, this method of generating the delay time can advantageously account for differences in user behavior between different restaurants. For example, users may leave their tables for longer periods of time at a buffet restaurant than at a restaurant with table service, and generating the delay time in this manner can account for such differences in behavior.

[0068] As another example, a gym behavior profile (e.g., for the gym smart zone **604B**) specifies a relatively long notification delay time (e.g., 2 hours) to reflect the tendency of people to leave their personal belongings in a locker or in their car before operating exercise equipment in the gym. Thus, if a user brings his or her mobile device **102** into the gym (e.g., to listen to music while operating exercise equipment), a tracking device **106** attached to a personal belonging left in a car or locker is likely to become disconnected for the duration of the user's workout. Similar to the restaurant example provided above, the delay time for a particular restaurant gym zone can be generated (or adjusted from a default value) based on tracking device data received from a plurality of users who have visited the gym.

[0069] As a third example, a church behavior profile specifies a notification delay time of 0 minutes (i.e., a notification is provided on the mobile device **102** as soon as a tracking device **106** has been disconnected) because it is relatively uncommon for people attending church services to become separated from their personal belongings. To reduce disruption to church services, a church behavior profile may further specify lower intensity values for the notification provided by the mobile device **102** (e.g., lower audio volume, lower vibration intensity, shorter vibration length, or generating a vibration in place of an audio alert) or the intensity of an alert provided by the tracking device **106** (e.g., a lower volume for an audio alert that the tracking device **106** plays back on a built-in speaker) upon detecting that the tracking device **106** has been disconnected.

[0070] In each of the example smart zones described above, the mobile device **102** may be configured to immediately display a notification to the user if the mobile device **102** leaves the smart zone while a tracking device **106** is disconnected, even if the notification delay time has not elapsed, because this means the user has departed from the smart zone and has left a personal belonging behind. For example, if a mobile device **102** leaves the gym smart zone **604B** after 15 minutes and the tracking device **106** attached to the user's house keys is not connected to the mobile device **102**, then the mobile device **102** displays a notification even though the 2-hour delay period has not elapsed. The modification of the behavior or configuration of a mobile device or tracking device when located in a smart zone is referred to herein as operation in a "smart zone mode".

[0071] The tracking system **100** may also maintain an unclassified behavior profile that the tracking system **100** assigns to smart zones that are not associated with a category or to smart zones that are associated to a category that does not have a category-specific behavior profile. The unclassified behavior profile may specify a default notification delay time (e.g., 1 hour) but may also be configured to immediately display a notification to the user if the mobile device **102** leaves the smart zone while a tracking device **106** is disconnected.

[0072] A third type of zone is a high-risk zone **606**. A high risk zone is a location in which a user has a higher-than-average probability of losing a personal belonging, such as an airport, train station, a concert venue, a sporting event, public transit, a store or other place of business, or a shopping mall. Similar to smart zones **604**, a high-risk zone **606** may also be associated with a behavior profile that specifies parameters governing the operation of the tracking device **106** and the associated features of the mobile device **102** when the user is present in the corresponding high-risk zone **606**. In general, a behavior profile for a high-risk zone may provide for more aggressive notifications, such as a shorter notification delay time (or a notification delay time of 0 minutes), higher-intensity notifications on the mobile device, greater transmission range for communications from the tracking device, a greater frequency of communications from the tracking device, and higher-intensity alerts on the tracking device.

[0073] High-risk zones can also be associated with a category representing a real world location, and each category can also be associated with a behavior profile specifying parameters that reflect the expected users in the corresponding type of location. For example, the high-risk zone **606** in FIG. **6** is an airport high-risk zone, and the tracking system **100** can define an airport behavior profile. In one embodiment, an airport behavior profile specifies a notification delay time of 0 minutes so that the mobile device **102** provides a notification as soon as a tracking device **106** is disconnected behavior of. Although this carries a risk of providing false positive notifications (i.e., providing a notification in a situation where a personal belonging has not actually been lost), the difficulty of recovering a lost item in an airport may be great enough to outweigh the inconvenience of providing false-positive notifications. An airport behavior profile may also specify a higher intensity for notifications provided on the mobile device **102** (e.g., higher volume, longer or more intense vibrations, a combination of vibration and audio alert) and for alerts provided on the tracking device **106** (e.g., a higher volume for audio alerts). An airport behavior profile may also configure the user's mobile device **102** to provide a notification when the separation distance of a tracking device **106** exceeds a threshold separation distance, such as 50 feet. This may be useful, for example, to alert the user when the user has left an item behind at a security checkpoint.

[0074] As another example, a behavior profile for a shopping mall may specify parameters that are more aggressive than average but less aggressive than an airport behavior profile because it is typically easier to recover lost personal belongings in a shopping mall than in an airport. For example, a shopping mall behavior profile may specify a notification delay time that is longer than 0 minutes but still relatively short (e.g., 5 to 10 minutes). The modification of the behavior or configuration of a mobile device or tracking device when present in a high risk zone is referred to herein as operation in a "high risk mode".

[0075] In some embodiments, different types of zones **602**, **604**, **606** can overlap. For example, a smart zone associated with a restaurant inside an airport terminal may overlap with a high-risk zone associated with the airport. As another, a safe zone associated with a user's home may overlap with a high-risk zone associated with an airport if the user lives near the airport. In one embodiment, a behavior profile associated with a safe zone is given higher priority than a behavior profile associated with a smart zone, which in turn is given higher priority than a behavior profile associated with a high-risk zone. Thus, if the mobile device **102** is in a location where a safe zone overlaps with a smart zone or a high-risk zone, the mobile device **102** and connected tracking device **106** are configured according to the behavior profile for the safe zone. Similarly, if the

mobile device **102** is in a location where a smart zone overlaps with a high-risk zone, the mobile device **102** and connected tracking device **106** are configured according to the behavior profile for the smart zone.

[0076] In some embodiments, a tracking device or a mobile device, once configured to operate in a special mode described herein (such as a safe zone mode, a smart zone mode, or a high risk mode), can be configured to operate in the special mode until the passage of a threshold amount of time. In other embodiments, the tracking device or mobile device can be configured to operate in the special mode until the central tracking system provides an instruction to operate in a different mode, until the tracking device or mobile device moves out of a corresponding zone), until the tracking device or mobile device detects movement of any kind (for instance, using an accelerometer or other motion-detection component), or based on any other suitable criteria being satisfied.

[0077] It should be noted that geographic boundaries associated with each type of zone described herein (safe zones, smart zones, and high risk zones, in addition to types of areas) can be defined based on a number of criteria. In some embodiments, the geographic boundaries are retrieved from an external source, for instance a database of property lines, boundaries from a map database, a location database maintained by the central tracking system, and the like. For example, for an airport high risk zone, the geographic boundaries can be the boundaries of the airport as defined by Google Maps. In other embodiments, the boundaries can be determined based on historical location data retrieved from a population of tracking devices, based on historical location data for a particular tracking device, or can be user-defined.

[0078] FIG. 7 illustrates a plurality of areas that each contain a plurality of zones, according to one embodiment. As referred to herein, an area is a region in which a user is expected to be located for a given time period. The example shown in FIG. 7 includes two different types of areas: a home area **700** and a travel area **710**.

[0079] A home area **700** is a geographic region in which the user resides. The tracking system **100** may generate the home area **700** for a user by drawing a circle of a predetermined radius (e.g., 50 miles) (or some other shape with predetermined dimensions) centered on the location of the user's home. In other embodiments, the home area **700** can be defined by the geographic boundary around the city, town, or neighborhood in which the user lives, or can be defined by the user. The user's home location may be provided as user input by the user, or the tracking system **100** may identify the user's home location by analyzing a location dataset for the user in the manner described below with reference to FIG. 8.

[0080] A travel area **710** is a geographic region in which the user is traveling. The tracking system **100** may generate a travel area **710** for a user based on the behavior of the user in a transit-related high-risk zone such as an airport, train station, or bus terminal. For example, if location information for the user (e.g., as recorded by the user's mobile device **102**) indicates that the user has entered and then exited a transit-related high-risk zone outside of his or her home area **700**, then the tracking system **100** generates a travel zone by drawing a circle of a predetermined radius (e.g., 50 miles) (or some other shape with predetermined dimensions) centered on the high-risk zone. The tracking system **100** may also generate a travel area based on the user's behavior at night when the user is outside of his or her home area. For example, if location information for the user indicates that the user's location remained stationary at a location associated with a hotel, then the tracking system generates a travel zone centered on the hotel. This method of generating a travel zone is advantageous, for example, because it can detect travel, such as a road trip, that does not involve passing through a transit-related high-risk zone such as an airport or train station. In an alternative embodiment, the tracking system **100** designates any geographic region beyond the user's home area as a travel area.

[0081] Each area is associated with a loss probability representing an overall probability that the user will lose a personal belonging while present in the area or an overall difficulty of recovering a personal belonging that has been lost in the area. In one embodiment, the tracking system **100**

generates the loss probability when generating the area. The loss probability can be generated based on the type of area with which the loss probability is associated (e.g., a lower loss probability is generated for a home area than for a travel area). The loss probability can also be based on other factors, such as the number of tracking devices **106** that were reported to the tracking system **100** as lost in the area in a preceding time period (e.g., the preceding 60 days).

[0082] When a user is present in an area, the tracking system **100** can configure the user's mobile device **102** and tracking device **106** based on the loss probability associated with the area. In one embodiment, the loss probability is stored as a loss coefficient (e.g., a coefficient greater than 1.0 represents a higher-than-average loss probability and a coefficient less than 1.0 represents a lower-than-average loss probability), and parameters of behavior profiles for smart zones and high-risk zones inside the area are scaled by the loss coefficient. For example, the travel area **710** shown in FIG. 7 has a loss coefficient of 1.25 to reflect that fact that it is generally more difficult to recover a personal belonging that has been lost while the user is traveling. As a result, the tracking system **100** scales the parameters of the behavior profiles for smart zones in the travel area **710** by the loss coefficient. For example, the notification delay times for the two restaurant smart zones in the travel area are reduced by a factor of 1.25 (i.e., divided by 1.25) to provide for prompter notifications when a tracking device **106** is disconnected. Similarly, the intensity of notifications (e.g., the volume of an audio alert or the intensity or duration of a vibration alert) is increased by a factor of 1.25 to provide for more intense notifications within the travel area **710**.

[0083] The tracking system **100** can also use the loss coefficient to modify the operation of the mobile device **102** and tracking device **106** independent of a behavior profile. For example, the default intensity of notifications provided on the mobile device **102** or alerts on the tracking device **106** is scaled by the loss coefficient when the user is present in the travel area **110**. As another example, the intensity or frequency of the beacon signal broadcast by the tracking device **106** is scaled by the loss coefficient when the user is present in the travel area **110**.

[0084] The tracking system **100** can also adjust other features of the tracking system **100**, mobile device **102**, or tracking device **106** based on the user's presence in an area. For example, if the user's mobile device **102** leaves the travel area **710** (e.g., when the user returns home from a vacation) but one of the user's tracking device **106** remains inside the travel area, then the tracking system **100** automatically designates the tracking device **106** as lost without receive a lost indication. In contrast, if the user's mobile device **102** leaves the home area **700** (e.g., when the user leaves home to go on a vacation) but one of the user's tracking device **106** remains inside the home area, then the tracking system **100** does not automatically designate the tracking device **106** as lost. Instead, the mobile device **102** may provide a notification to the user to alert the user that the object attached to the tracking device **106** has been left behind.

[0085] FIG. 8 is a flow chart illustrating a process **800** for generating zones, according to one embodiment. In other embodiments, the process **800** may include additional steps not shown in FIG. 8, and some of the steps in the process **800** may be omitted or performed in a different order. Although the description provided with reference to FIG. 8 states that the process **800** is performed on the tracking system **100**, some or all of the steps in the process may alternatively be performed on the mobile device **102**.

[0086] The tracking system **100** receives **810** a location dataset for the user. A location dataset comprises a set of timestamped locations representing the places where the user has traveled during a trailing time window (e.g., the preceding 30 days or the preceding 60 days). As referred to herein, a location point is a single timestamped location (e.g., a latitude-longitude coordinate pair associated with a date and time), and the location dataset comprises a plurality of location points. In one embodiment, the location points are recorded by the user's mobile device **102** and sent to the tracking system **100**. For example, a user's mobile device **102** is configured to periodically record the location of the mobile device **102** as indicated by a location tracking function on the mobile device **102** (e.g., a GPS receiver on the mobile device **102**), and this location is presumed to

represent the user. In one embodiment, the mobile device **102** provide privacy settings that allow the user to prevent the mobile device **102** from recording location points and/or sending the location points to the tracking system.

[0087] The tracking system **100** generates **812** safe zones by identifying the densest locations in the location dataset. In some embodiments, the tracking system **100** identifies the densest locations in the location dataset based on a clustering algorithm that groups densely connected location points into clusters. The tracking system **100** may perform the clustering algorithm on the entire location dataset or on a subset of the location points in the location dataset, such as the subset of location points in a shorter trailing time window. For example, the tracking system maintains a location dataset that includes location points from the preceding 60 days, and the clustering algorithm is performed on the location points from the preceding 30 days. The tracking **100** may also perform one or more data preprocessing and filtering steps on the location points before performing the clustering algorithm. For example, the tracking system **100** applies an accuracy filter to the location points to remove location points whose accuracy is less than a threshold (e.g., removing location points whose error is greater than ± 65 meters).

[0088] The clustering algorithm may be configured to identify clusters that include a number of location points exceeding a clustering threshold, whereas groups that include a number of location points below the clustering threshold are not grouped into clusters and are instead identified as outliers. For example, the clustering threshold may be set to 50 location points, which means groups containing more 50 or more location points are joined into a cluster, whereas location points in groups containing fewer than 50 location points are identified as outliers. As another example, the clustering threshold may be set to 20 location points. In one embodiment, the clustering algorithm is a density-based spatial clustering of applications with noise (DBSCAN) algorithm.

[0089] After identifying the clusters, the tracking system **100** generates **812** safe zones at some or all of the clusters. In one embodiment, the tracking system **100** generates **812** a safe zone at each cluster identified by the clustering algorithm. In another embodiment, the tracking system **100** selects a subset of the clusters and generates **812** a safe zone at each selected cluster. For example, the tracking system **100** assigns a density value to each cluster (e.g., the density value may be the number of location points in the cluster). The tracking system **100** may then rank the clusters according to their density values and select the clusters whose ranking exceeds a threshold ranking (e.g., the five densest clusters) or select the clusters whose density value exceeds a threshold density value (e.g., the clusters having density value exceeding 75 location points). It should be noted that in some embodiments, safe zones are user defined, while in other embodiments, candidate safe zones are presented to a user for confirmation as safe zones. For instance, the tracking system **100** can generate a set of candidate safe zones based on the identified clusters, can suggest that a user affirmatively select one or more of the set of candidate safe zones, and can limit the generate safe zones to the selected candidate safe zones.

[0090] After generating **812** the safe zones, the tracking system **100** can categorize the safe zones. In one embodiment, the tracking system **100** categorizes a safe zone based on the timestamps of the location points in the safe zone. For example, a user's location tends to be the same as the location of his or her home between 12 AM and 6 AM (i.e., when the user is sleeping). Thus, if the timestamps of the location points in a particular safe zone include a large number of timestamps between 12 AM and 6 AM, then the safe zone may be categorized as the user's home safe zone. Similarly, a user's location tends to be the same as the location of his or her workplace between 9 AM and 5 PM, so if the timestamps of the location points in a particular safe zone include a large number of timestamps between those hours, then the safe zone may be categorized as the user's workplace safe zone.

[0091] The tracking system **100** identifies **814** high-risk zones in the region. As noted above, high-risk zones can be applicable to multiple users, whereas smart zones and safe zones may be specific to a single user. As a result, high-risk zones may be generated in a process independent of the

process **800** shown in FIG. **8**. In one embodiment, the process determines whether to identify a location based on the number of lost indications received at a location, the number of found indications received at the same location, and whether publicly available location or map data indicates that the location has a category that is part of a predetermined list of high-risk categories maintained on the tracking system **100** (e.g., an airport, helipad, airbase, train station, bus terminal, concert venue, shopping mall, etc.). For example, the process computes a found-to-lost ratio representing the ratio of found indications to lost indications at a location. If the ratio is less than a threshold (e.g., there are relatively few found indications relative to lost indications at the location), then the process determines whether the location has a high-risk category. If the location has both a ratio below the threshold and a high-risk category, then the tracking system **100** generates a high-risk zone at the location.

[0092] The tracking system **100** generates **816** smart zones by identifying dense locations in the location dataset and filtering out the high-risk zones and the safe zones. In one embodiment, the tracking system **100** performs the clustering algorithm a second time, but with a lower clustering threshold (e.g., the clustering threshold is set to 50 to generate **812** safe zones and set to 20 to generate **816** smart zones), and then removes any clusters that have already been used to generate safe zones and any clusters at locations where a high-risk zone has been generated. The tracking system **100** then generates **816** smart zones at the clusters that were not removed.

[0093] In addition to a different clustering threshold, the tracking system **100** may also run the clustering algorithm on a different subset of the location dataset when generating **816** the smart zones. For example, the tracking system **100** provides location points from the preceding 30 days as input to the clustering algorithm when generating **812** the safe zones and provides location points from the preceding 60 days as input to the clustering algorithm when generating **816** the smart zones. The tracking system **100** may additionally perform one or more data preprocessing and filtering steps on the location points (e.g., removing location points whose accuracy is lower than a threshold) before performing the clustering algorithm to generate **816** the smart zones.

[0094] After generating **816** the smart zones, the tracking system **100** may classify some or all of the smart zones with a smart zone category (e.g., the restaurant smart zone, gym smart zone, and church smart zone as described above with reference to FIG. **6**). For example, the tracking system **100** receives information about the location of the smart zone from a map database (e.g., via a call to the GOOGLE PLACES API) and classifies the smart zone based on the location information.

[0095] The tracking system **100** also generates a boundary for each smart zone. The boundary for a smart zone can be generated, for example, based on the locations of location points in the cluster (e.g., the boundary covers a larger area if the location points are farther apart from each other), a predetermined radius (e.g., the boundary is a circle with the predetermined radius and centered on the center of the cluster), property data (e.g., boundaries of the plot of land containing the location, according to map data), or an analysis of past users' behavior in at the location (e.g., the average amount of movement of users within the location, the average rate at which users enter and leave the location, etc.). In one embodiment, the tracking system **100** stores the boundary as a geofence. The tracking system **100** can also perform a similar process to generate boundaries for the safe zones generated **812** by the tracking system **100**.

[0096] The tracking system **100** creates **818** behavior profiles for each of the generated zones. As noted above with reference to FIG. **6**, the tracking system **100** can generate the parameters in the behavior profile for a smart zone based on the category of the smart zone and based on tracking device data received from a plurality of users who have visited the smart zone. The tracking system **100** may also create **818** behavior profiles for the safe zones.

[0097] The tracking system **100** stores **820** the created zones and behavior profiles. For example, the zones and behavior profiles are stored in the zone database **218** or in the user database **210** in association with the user's other data.

[0098] In addition to the process **800** described above, the tracking system **100** may also generate

smart zones with other methods. In one embodiment, the tracking system **100** also performs a process that can generate a temporary smart zone for a public event such as an outdoor concert or a parade. For example, the tracking system **100** monitors public places where public events typically occur (e.g., the National Mall in Washington, D.C.) and creates a temporary smart zone in the public place when the number of people in the public place exceeds a threshold number. The threshold number can be determined based on an average number of people in the public place (e.g., the threshold number is three times the average number, so a temporary smart zone is created when three times as many people as average are in the public place. The tracking system **100** can determine the number of people in the public place, for example, based on location points received from mobile devices **102** in the public place over a relatively short trailing time window (e.g., the preceding hour). The tracking system **100** can additionally or alternatively generate a temporary smart zone based on publicly available information that provides a time and place for a public event or by user input from an operator of the tracking system **100** or a user of mobile device **102** in communication with the tracking system.

[0099] The tracking system **100** may also generate smart zones that are applicable to multiple users (similar to the high-risk zones described above) by combining location datasets for multiple users and providing the combined location dataset to the clustering algorithm described above. The tracking system **100** can then generate **816** the smart zones in the same manner as described above by performing the clustering algorithm to identify dense locations and then filtering out high-risk zones and safe zones. However, rather than filtering out the safe zones for a single user, the tracking system **100** filters out the safe zones for each of the multiple users whose location datasets were provided as input to the clustering algorithm.

[0100] FIG. **9** is a flow chart illustrating a process **900** for modifying tracking device behavior based on presence in a smart zone, according to one embodiment. In other embodiments, the process **900** may include additional steps not shown in FIG. **8**, and some of the steps in the process **900** may be omitted or performed in a different order.

[0101] The mobile device **102** detects **910** the user's presence in the smart zone. For example, location data from a location tracking feature on the mobile device **910** indicates that the mobile device **102** is inside a geofence that defines the boundaries of a smart zone. After detecting **910** the user's presence in the smart zone, the mobile device **102** accesses the behavior profile for the smart zone. For example, the mobile device **102** retrieves the behavior profile from the tracking system **100**. As another example, the mobile device **102** maintains a local cache of behavior profiles for frequently visited smart zones, and the mobile device **102** accesses the behavior profile in the cache.

[0102] The mobile device **102** reconfigures **912** a connected tracking device **106** based on the behavior profile. For example, the behavior profile may specify parameters that change settings on the tracking device **106**, such as the power and frequency of the beacon signal broadcast by the tracking device **106** or the volume of audio alerts emitted by a speaker on the tracking device **106**, and the mobile device **102** reconfigures **912** the tracking device **106** by applying these settings. The mobile device **102** also reconfigures **914** itself based on the behavior profile. For example, the mobile device **102** adjusts its notification delay time to match the notification delay time specified in the behavior profile. The mobile device **102** may also adjust the intensity of the notifications that it provides based on parameters specified in the behavior model.

[0103] FIG. **10** is a flow chart illustrating a process **1000** for modifying tracking device behavior based on presence in an area, according to one embodiment. In other embodiments, the process **1000** may include additional steps not shown in FIG. **8**, and some of the steps in the process **1000** may be omitted or performed in a different order.

[0104] The mobile device **102** detects **1010** the user's presence in an area. For example, in an embodiment where the home area is defined as a circle centered on the user's home with a predetermined radius (e.g., 50 miles), then the mobile device **102** determines a distance between

the location of the mobile device **102** and the location of the user's home. If the distance is less than the predetermined radius, then the mobile device **102** is inside the home area. Similarly, if a travel area is defined as a circle with a predetermined radius centered on a transit-related high-risk zone or a location associated with a hotel, the mobile device **102** determines that it is inside the travel area if the distance between the mobile device and the center of the travel area is less than the predetermined radius.

[0105] The mobile device **102** reconfigures **1012** a connected tracking device **106** based on the loss probability for the area. The mobile device **102** also reconfigures **1014** itself based on the loss probability for the area. For example, the mobile device **102** may modify its own settings or settings of the tracking device **106** based on a loss coefficient in the manner described above with reference to FIG. 7.

[0106] Although the steps of the processes **900**, **1000** shown in FIGS. **9** and **10** are described as being performed on the mobile device **102**, some or all of the steps of these processes **900**, **1000** may alternatively be performed on the tracking system **100**.

Additional Considerations

[0107] The foregoing description of the embodiments of the invention has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

[0108] Any of the devices or systems described herein can be implemented by one or more computing devices. A computing device can include a processor, a memory, a storage device, an I/O interface, and a communication interface, which may be communicatively coupled by way of communication infrastructure. Additional or alternative components may be used in other embodiments. In particular embodiments, a processor includes hardware for executing computer program instructions by retrieving the instructions from an internal register, an internal cache, or other memory or storage device, and decoding and executing them. The memory can be used for storing data or instructions for execution by the processor. The memory can be any suitable storage mechanism, such as RAM, ROM, flash memory, solid state memory, and the like. The storage device can store data or computer instructions, and can include a hard disk drive, flash memory, an optical disc, or any other suitable storage device. The I/O interface allows a user to interact with the computing device, and can include a mouse, keypad, keyboard, touch screen interface, and the like. The communication interface can include hardware, software, or a combination of both, and can provide one or more interfaces for communication with other devices or entities.

[0109] Some portions of this description describe the embodiments of the invention in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

[0110] Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

[0111] Embodiments of the invention may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may

comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

[0112] Embodiments of the invention may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

[0113] Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the invention be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments of the invention is intended to be illustrative, but not limiting, of the scope of the invention, which is set forth in the following claims.

Claims

1. A method for locating a tracking device, comprising: receiving, by a central tracking system, a location of a tracking device coupled to a pet of a tracking device owner; querying, by the central tracking system, a zone database to identify one or more zones corresponding to the received location, each zone including a geographic boundary; in response to determining that an identified zone in which the tracking device coupled to the pet is location is a safe zone, flagging, by the central tracking system, the tracking device coupled to the pet as safe; and in response to determining that the identified zone is a high-risk zone and that the tracking device coupled to the pet is not within communicative range of a device of the tracking device owner, flagging, by the central tracking system, the tracking device coupled to the pet as lost and sending a notification to the tracking device owner indicating that the tracking device coupled to the pet is lost.
2. The method of claim 1, wherein a first geographic boundary corresponds to a home of the user.
3. The method of claim 1, wherein a first geographic boundary corresponds to a city, town, or neighborhood in which the user lives.
4. The method of claim 1, wherein a first geographic boundary is defined by the user.
5. The method of claim 1, wherein a signal transmission rate of the tracking device is selected based on whether the tracking device is flagged as safe.
6. The method of claim 1, wherein a signal transmission power of the tracking device is selected based on whether the tracking device is flagged as safe.
7. The method of claim 1, wherein a configuration of the tracking device is selected based on whether the tracking device is flagged as safe.
8. A non-transitory computer-readable storage medium storing executable computer instructions that, when executed by a processor, perform steps comprising: receiving, by a central tracking system, a location of a tracking device coupled to a pet of a tracking device owner; querying, by the central tracking system, a zone database to identify one or more zones corresponding to the received location, each zone including a geographic boundary; in response to determining that an identified zone in which the tracking device coupled to the pet is location is a safe zone, flagging, by the central tracking system, the tracking device coupled to the pet as safe; and in response to determining that the identified zone is a high-risk zone and that the tracking device coupled to the pet is not within communicative range of a device of the tracking device owner, flagging, by the central tracking system, the tracking device coupled to the pet as lost and sending a notification to

the tracking device owner indicating that the tracking device coupled to the pet is lost.

9. The non-transitory computer-readable storage medium of claim 8, wherein a first geographic boundary corresponds to a home of the user.

10. The non-transitory computer-readable storage medium of claim 8, wherein a first geographic boundary corresponds to a city, town, or neighborhood in which the user lives.

11. The non-transitory computer-readable storage medium of claim 8, wherein a first geographic boundary is defined by the user.

12. The non-transitory computer-readable storage medium of claim 8, wherein a signal transmission rate of the tracking device is selected based on whether the tracking device is flagged as safe.

13. The non-transitory computer-readable storage medium of claim 8, wherein a signal transmission power of the tracking device is selected based on whether the tracking device is flagged as safe.

14. The non-transitory computer-readable storage medium of claim 8, wherein a configuration of the tracking device is selected based on whether the tracking device is flagged as safe.

15. A system comprising a hardware processor and a non-transitory computer-readable storage medium storing executable instructions that, when executed by the hardware processor, cause the system to perform steps comprising: receiving, by the system, a location of a tracking device coupled to a pet of a tracking device owner; querying, by the system, a zone database to identify one or more zones corresponding to the received location, each zone including a geographic boundary; in response to determining that an identified zone in which the tracking device coupled to the pet is location is a safe zone, flagging, by the system, the tracking device coupled to the pet as safe; and in response to determining that the identified zone is a high-risk zone and that the tracking device coupled to the pet is not within communicative range of a device of the tracking device owner, flagging, by the system, the tracking device coupled to the pet as lost and sending a notification to the tracking device owner indicating that the tracking device coupled to the pet is lost.

16. The system of claim 15, wherein a first geographic boundary corresponds to a home of the user.

17. The system of claim 15, wherein a first geographic boundary corresponds to a city, town, or neighborhood in which the user lives.

18. The system of claim 15, wherein a first geographic boundary is defined by the user.

19. The system of claim 15, wherein a signal transmission rate of the tracking device is selected based on whether the tracking device is flagged as safe.

20. The system of claim 15, wherein a signal transmission power of the tracking device is selected based on whether the tracking device is flagged as safe.
